World Journal of Acupuncture – Moxibustion 32 (2022) 131–136

Contents lists available at ScienceDirect

World Journal of Acupuncture – Moxibustion

journal homepage: www.elsevier.com/locate/wjam

Clinical Research

Electroacupuncture on adjuvant analgesia after total knee arthroplasty: A randomized controlled trail☆

电针对全膝关节置换术后辅助镇痛的临床疗效观察

Wei SHI (石威)[a], Yan HOU (侯燕)[b], Wei YAN (严伟)[b], Yong-hai TAN (谭勇海)[b], Xin YU (余昕)[b], Lei LI (李磊)[b], De-bao ZOU (邹德宝)[b], Jie WANG (王杰)[b], Zhi-zhou WANG (王志洲)[a], Hong-jiang JIANG (姜红江)[a,b,*]

[a] Graduate School of Anhui University of Chinese Medicine, Hefei, Anhui Province 230000, China (安徽中医药大学研究生院, 合肥230000, 中国)

[b] Third Department of Arthropathy, Shandong Wendeng Osteopathy Hospital, Weihai, Shandong Province 264400, China (山东省文登整骨医院关节三科, 威海264400, 中国)

A R T I C L E I N F O

Article history:
Available online 9 March 2022

Keywords:
Total knee arthroplasty
Postoperative pain
Electroacupuncture
Analgesia
Effect mechanism

A B S T R A C T

Objective: To observe the clinical effect of electroacupuncture (EA) for alleviating pain after total knee arthroplasty (TKA) in the patients with the end-stage knee osteoarthritis and its mechanism.
Methods: A total of 80 subjects with unilateral TKA were divided into a celecox group (40 cases) and an EA combination group, using random number table in clinical trial. In the celecox group, after operation, celecox tablets were taken orally for analgesia. In the EA combination group, on the base of oral administration of celecox tablets, on the day of operation, EA was exerted on the affected limb, with disperse-dense wave, 2 Hz/100 Hz in frequency, 3 mA in current intensity, once daily, 30 min each time, consecutively for 3 days. On the day after operation, Wàiqiū (外丘GB36), Jīnmén (金门BL63), Nèimádiǎn (内麻点EX-LE29) and Zúsānlǐ (足三里ST36) were selected in EA. 24 h after removing the drainage tube, Liángqiū (梁丘ST34), Xuèhǎi (血海SP10), Dìjī (地机SP8) and Yánglíngquán (阳陵泉GB34) were used. Separately, the operation time, blood loss, the pain VAS score before operation, the pain VAS scores after 24, 48 and 72 h of treatment and Self-rating scale of sleep (SRSS) score, as well as the stress factors, i.e. IL-6, IL-10 and TNF-α in serum were recorded and determined in the subjects of two groups.
Results: Before treatment, the operation time and blood loss were not different between two groups (both $P > 0.05$). 24, 48 and 72 h after operation, the visual analogue scale (VAS) scores in the celecox group were all higher than those in the EA combination group (all $P < 0.05$). After treatment, SRSS score in the EA combination group was lower than that in the celecox group ($P < 0.05$). The expressions of IL-6 and TNF-α were reduced gradually in the two groups (all $P < 0.05$) and the values in the EA combination group were lower than the celecox group ($P < 0.05$). IL-10 was increased gradually ($P < 0.05$) and the value in the EA combination group was higher than the celecox group ($P < 0.05$).
Conclusion: Electroacupuncture effectively alleviates pain induced by TKA and the curative effect is definite. This therapy contributes to accelerating rehabilitation and its effect mechanism may be related to promoting the release of anti-inflammatory factor, IL-10 and inhibiting the expressions of inflammatory factors, IL-6 and TNF-α.

Total knee arthroplasty (TKA) is the most effective and mature treatment for end-stage knee osteoarthritis to reconstruct knee function and relieve knee pain in patients [1]. However, most of patients suffer from pain to various degrees after TKA. On the one hand, postoperative pain seriously affects the early function rehabilitation of knee joints, which leads to deep vein thrombosis and joint stiffness. On the other hand, postoperative pain causes stress response, resulting in anxiety, sleep disorder and reduced quality of life. If the postoperative pain cannot be treated in time and effectively, the adverse effects may be induced in physiological systems, e.g. circulatory system and respiratory system [2,3]. With the accelerated development of surgical rehabilitation, the anal-

☆ Supported by the Fourth Batch Key TCM Specialty Construction Projects in Weihai City (Orthopedics) (Weihai Health Administration Office [2019] No. 87).

* Corresponding author at: Third Department of Arthropathy, Shandong Wendeng Osteopathy Hospital, Weihai, Shandong Province 264400, China (山东省文登整骨医院关节三科, 威海264400, 中国)

E-mail address: boneman@163.com (H.-j. JIANG).

https://doi.org/10.1016/j.wjam.2022.03.003
1003-5257/

gesic method after TKA has become a hot topic of clinical research. Currently, the pain management principle of TKA runs throughout the preoperative, intraoperative and post-operative treatment. Of them, the postoperative pain management includes postoperative prophylactic analgesia and pain treatment. The prophylactic analgesic method is adopted as the top option. If the score of visual analogue scale (VAS) is $\geq$ 3 points, pain treatment should be selected immediately [4].

Electroacupuncture (EA) is developed on the basis of acupuncture and it sustainably stimulates the body, is conductive to enhancing the strength of *deqi*, promoting *qi* and blood circulation, regulating *yin* and *yang*, removing the obstruction in meridians and stopping pain. It has been proved that EA is advantageous at quick onset of effect and absence of toxic and side effects in treatment of pain disorders. Studies have shown [5,6] that EA can effectively relieve postoperative pain and have few adverse reactions. However, there is less research of the mechanism of EA assisted on analgesia after TKA and the impacts on stress factors. In this research, based on the meridian theory of traditional Chinese medicine (TCM), EA therapy was adopted to observe its adjuvant effect of analgesia after TKA in clinic. Through observing the expressions of IL-6, IL-10 and TNF-α in serum, the effect mechanism of EA was explored on adjuvant analgesia after TKA. The results are reported as follows.

Data and methods

Clinical data

The patients with unilateral TKA were selected as the subjects from Department of Arthropathy, Shandong Wendeng Bonesetting Hospital from May 2019 through October 2020 and a total of 80 cases were included in clinical trial. In order to avoid selection bias, a specially-assigned person used random number table method for inclusion and group division through computer. Finally, a celecox group and an EA combination group were divided, 40 cases in each one. This trial had been approved by Medical Ethics Committee of Shandong Wendeng Osteopathy Hospital (Ethics Approval Number: No. 2019-015).

Diagnosis criteria

Diagnosis criteria were in reference to the clinical and radiological diagnosis criteria of knee osteoarthritis recorded in *Guidelines for osteoarthritis (2018)* [7].

Inclusion criteria

(1) In compliance with above diagnosis criteria and having accepted TKA. (2) Aged 55–75 years, either sex. (3) Patients suffering from end-stage knee osteoarthritis and having accepted TKA. (4) Participated in the trial voluntarily, fully understanding of the trail procedure by the families and signed informed consent form.

Exclusion criteria

(1) Accepted joint cavity drug injection within nearly 6 months before TKA. (2) Suffering from serious neurological diseases and complicated autonomous basic diseases, e.g. cardiocerebral vascular disorders and liver and kidney insufficiency. (3) Suffering from acute or chronic suppurative knee joint infection before operation. (4) Suffering from nerve and vessel damages on the affected limb before operation. (5) Having taken aspirin enteric tablets for nearly 1 week before operation.

Dropout criteria

(1) Suffering from serious nerve injury and deep vein thrombosis after TKA. (2) Failure to judge the therapeutic effect after operation. (3) Incomplete follow-up data. (4) Prosthetic infection after operation and requiring to have a repair.

Withdrawal criteria

(1) Having a strong request of withdrawal during treatment. (2) Being afraid of acupuncture or occurring fainting needle. (3) Intolerance of postoperative pain and required analgetic for many times. (4) Death.

Methods

Operation was performed under epidural anesthesia. It was operated by the same senior orthopedic surgeon and participated by the same team. The longitudinal incision in the middle of the anterior knee and the parapatellar medial access were adopted. From 10 cm above the patella to 2 cm inferior to the medial tibial nodule, the incision was exerted through skin, subcutaneous tissue and deep fascia successively, and then, hyperplastic osteonsoma and synovial membrane were removed. Afterwards, the osteotomy of the distal femur and proximal tibia were performed using an intramedullary positioning guide rod and an extramedullary locator respectively. After commissioning, the knee joint prosthesis was installed. In both groups, tranexamic acid, 100 mL was injected after incision closure and the medication with antibiotics and anticoagulant drug was administered after operation. During operation, the patella was not replaced. The implants were the surface knee joint prosthesis of the same type, Beijing Chunli brand, made in China.

In the celecox group, after operation, celecox tablets were taken orally (200 mg/time, twice a day) for analgesia.

In the EA combination group, on the base of oral administration of celecox tablets for analgesia, on the day of operation, EA was performed on the affected limb, with the predominant disperse-dense wave for analgesia [8], 2 Hz/100 Hz in frequency, 3 mA in current intensity, once daily, 30 min each time, consecutively for 3 days. On the day after treatment, in order to reduce surgical infection, the distal points were selected, including Wàiqiū (‘¤‘–‘¸‘˜GB36), Jīnmén (金门BL63), Nèimádiăn (内麻点EX-LE29) (located on the medial side of the leg, at the middle of the line between the medial condyle of the tibia and the top of the medial malleolus, 1 finger-breath posterior to the medial side of the tibia) [9] and Zúsānlĭ (‘¶‘³‘¸‘‰‘‡‘ŒST36). One pair of electrodes were attached to GB36 and EX-LE29 and another one to ST36 and BL63. On the 2nd day of operation, Liángqiū (‘¢‘"‘¸‘˜ST34), Xuèhăi (‘¡‘Ł‘µ‘·SP10), Dìjī (‘œ‘°‘œ‘ºSP8) and Yánglíngquán (‘˜‘³‘™‘µ‘³‘‰GB34) were selected. One pair of electrodes were connected at SP8 and SP10 and another one at ST34 and GB34. Acupuncture was operated by the same acupuncture physician, with the needle (*Hwato* 0.25 mm $\times$ 40 mm) and the electric treatment apparatus (Great Wall KWD-808 I) of the same brand.

Observation indicators and determination methods

The pain VAS score

It is a 10 cm long ruler ranging from 0 (no pain) to 10 (intolerable pain). Successively, before operation, 24, 48 and 72 h after operation, patients chose the corresponding range on the ruler according to their own pain sensation. The data collector determined the corresponding score based on the indicated location of the ruler.

Table 1
Comparison of general data in the subjects with TKA between two groups (Mean ± SD).

Groups	Patients	Age (years)	Sex (cases)		Affected knee (cases)		Operation time (min)	Blood loss (mL)
			M	F	L	R		
Celecox	40	67.0 ± 4.9	16	24	21	19	93.9 ± 4.0	413.1 ± 10.2
EA combination	40	66.7 ± 4.6	18	22	23	17	94.3 ± 4.0	411.4 ± 12.6

Self-rating scale of sleep (SRSS) score

In 24, 48 and 72 h after operation, SPSS score was collected successively. Using this international universal scale, the score was determined in compliance with sleep state and sleep factors. Total score < 4 points: no sleep disturbance. Total score ranged from 4–6 points: sleep disturbance. Total score > 6 points: insomnia.

Detection of stress factors in serum, i.e. IL-6, IL-10 and TNF-α

24, 48 and 72 h after operation, the venous blood sample, 5 mL was collected. After centrifugation at 3000 r/min for 30 min, using enzyme-linked immunosorbent assay, the levels of IL-6, IL-10 and TNF-α in serum were detected. The kit was from Biyuntian Biological Kit Company and the operation was strictly in accordance with the kit instruction.

Statistical management

SPSS 23.0 was performed for all data analysis. Mean ± standard deviation (mean± *SD*) was used to express measurement data, the *t*-test was adopted if the data were in compliance with the normal distribution. Enumeration data were expressed with rate or percentage and analyzed with χ^2 test. Significant level was expressed as $\alpha = 0.05$ and $P < 0.05$ indicated statistical difference.

Results

General data

In this trial, 80 subjects were finally included in the observation of clinical therapeutic effect, of which, there were 34 males and 46 females, 44 cases with left knee involved and 36 cases with the right knees. All of the subjects were successful in operation and actively cooperated with treatment. The drainage tube was removed 24 h after operation and the patients could have the function exercise positively and have a walk under the auxiliary device 1 week after operation. 2 weeks after operation, the suture was taken out and the incision healing was of A grade. No case was dropped out or withdrawn due to complications, such as infection, neurovascular injury and prosthesis loosening and displacement. The postoperative follow-up was successful and the data of 40 subjects of each group were included in analysis. There were not statistical differences in the general data in the patients between two groups, e.g. age, sex, affected knee, operation time and blood loss (intraoperative blood loss + drainage amount 24 h after operation) ($P > 0.05$), indicating the comparability. See Table 1.

Comparison of VAS score in the subjects between two groups

Illustrated in Fig. 1 and Table 2, VAS score was not different statistically before operation in the subjects between two groups (all $P > 0.05$), suggesting the comparability. 24, 48 and 72 h after operation, VAS scores were different statistically between two groups (all $P < 0.05$).

In the celecox group, as compared with the score before operation, 24, 48 and 72 h after operation, the scores were different statistically (all $P < 0.05$). In the EA combination group, 48 and 72 h after treatment, the scores were different statistically as compared with those before operation separately (both $P < 0.05$).

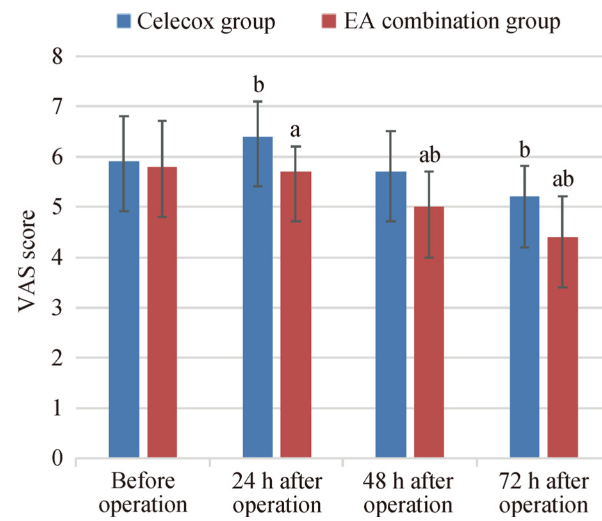

Fig. 1. Comparison of VAS score after TKA in the subjects between two groups (Mean ± SD, score).
Notes:
[a]Compared with the celecox group, $P < 0.05$.
[b]Compared with before operation in either group, $P < 0.05$.

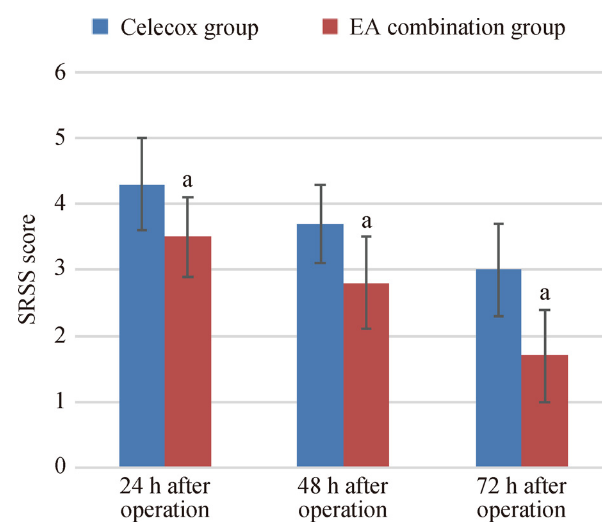

Fig. 2. Comparison of SRSS score after TKA in the subjects between two groups (Mean ± SD, score)
Note: [a]Compared with the celecox group, $P < 0.05$.

Comparison of postoperative SRSS score in the subjects between two groups

The comparison of SRSS score in the subjects between two groups in 24, 48 and 72 h of operation is illustrated in Fig. 2 and Table 3. There were statistical differences in SRSS scores 24, 48 and 72 h after operation between groups successively (all $P < 0.05$).

Table 2
Comparison of the pain VAS score in the subjects after TKA between two groups (Mean ± SD, score).

Groups	Patents	Before operation	24 h after operation	48 h after operation	72 h after operation
Celecox	40	5.9 ± 0.9	6.4 ± 0.7[b]	5.7 ± 0.8	5.2 ± 0.6[b]
EA combination	40	5.8 ± 0.9	5.7 ± 0.5[a]	5.0 ± 0.7[a,b]	4.4 ± 0.8[a,b]

Notes:.
[a] Compared with the celecox group, $P < 0.05$.
[b] Compared with before operation, $P < 0.05$.

Table 3
Comparison of SRSS score in the subjects after TKA between two groups (Mean ± SD, score).

Groups	Patients	24 h after operation	48 h after operation	72 h after operation
Celecox	40	4.3 ± 0.7	3.7 ± 0.6	3 ± 0.7
EA combination	40	3.5 ± 0.6 [a]	2.8 ± 0.7 [a]	1.7 ± 0.7[a]

Note: [a]Compared with the celecox group, $P < 0.05$.

Table 4
Comparison of stress factors in the subjects after TKA between two groups (Mean ± SD).

Groups	Patients	Time	IL-6	IL-10	TNF-α
Celecox	40	24 h after operation	55.3 ± 3.8	42.3 ± 1.3	84.9 ± 2.6
		48 h after operation	36.7 ± 1.3	50.9 ± 1.7	84.9 ± 2.6
		72 h after operation	11.7 ± 1.8	56.0 ± 1.5	44.2 ± 5.1
EA combination	40	24 h after operation	46.0 ± 3.1[a]	45.2 ± 1.7[a]	75.9 ± 2.0[a]
		48 h after operation	27.4 ± 1.8[a]	55.7 ± 1.4[a]	75.9 ± 2.0[a]
		72 h after operation	7.2 ± 1.6[a]	62.4 ± 1.4[a]	34.1 ± 2.6[a]

Note: [a]Compared with the celecox group, $P < 0.05$.

Comparison of IL-6, IL-10 and TNF-α in the subjects between two groups

The comparison of stress factors in the subjects between two groups in 24, 48 and 72 h of operation is shown in Table 4. IL-6 and TNF-α were reduced gradually in two groups (all $P < 0.05$) and those two factors in the EA combination group were lower than the celecox group (both $P < 0.05$). IL-10 was increased gradually in the two groups (both $P < 0.05$) and the level of it in the EA combination group was higher than that in the celecox group ($P < 0.05$).

Typical case

Wang XX, female, 71 years old, suffering from swelling and pain of both knee joints and motor impairment for 10 years. Digital X-ray film showed that there were sharp-formed hyperosteogeny around the joints and intercondylar protrusion of both knees, narrowing joint space and joint sclerosis as well as hyperplasia and sclerosis found on the joint surface of both patellas. TKA was exerted and postoperative EA was assisted for analgesia. See Fig. 3.

Discussion

Osteoarthritis is one of the most common diseases and causes disability in elderly, which is related to sex, age, body mass index, lower limb mechanical axis and biomechanics [10–12]. Its lesion is mainly located in the joints with load-bearing function, i.e. hip, knee and ankle joints. Of those joints, the incidence of knee joint is the highest, manifested in clinic as degeneration and abrasion of cartilage and subchondral bone [13,14]. TKA is the most effective approach to the treatment of knee osteoarthritis. This therapy alleviates pain, corrects deformity, restores the lower extremity force line and reconstructs joint function so as to improve the quality of life in patients. With the social aging, the demand for TKA increases significantly. The study [15] indicates that the survival rate is 95% approximately in the patients with TKA. Therefore, it is crucial to effectively and quickly relieve postoperative pain and thus improve patients' satisfaction.

In TCM, the pathogenesis of pain includes no more than two aspects, namely where is obstruction, where is pain, and where is malnutrition, where is pain. The former refers to pain induced by the stagnation and obstruction of *qi*, blood and meridians. The latter explains that pain is related to malnutrition of meridian, which is caused by *qi* and blood insufficiency. The mechanism of acupuncture analgesia has a long history in TCM. It has been early proposed in *Língshū* (《灵枢》*Miraculous Pivot*) that "the rule in applying acupuncture lies in knowing how to regulate *yin* and *yang*. Only when *yin* and *yang* are regulated, can essence *qi* be replenished, the body and *qi* be integrated and the spirit maintained in side". It means that *qi* and blood is harmonized only when *yin* and *yang* are balanced. It is clearly indicated in Chapter 75 of *Língshū* (《灵枢》*Miraculous Pivot*) that "the key using acupuncture therapy to treat disease lies in the regulation of *qi*". In Chapter 1 of *Língshū* (《灵枢》*Miraculous Pivot*), it is said that "the arrival of *qi* ensures curative effect, and the significant curative effect appears like wind blowing away clouds". It is clear that acupuncture is effective in analgesia, depending on "*qi* regulation" [16]. Whenever *qi* is of free flow, blood circulation is normal. After TKA, meridians are damaged, thus, stasis is remained in the local of knee joints, resulting in swelling and pain in patients. This postoperative disorder refers to "tendon *bi* syndrome". On the base of the theory mentioned above, in association with the idea as "distal acupoints acting on spectral effect, while local ones on segmental effect" [17], five principles of acupoint selection were adopted in this trial. (1) Selecting the acupoints with significant effect on acute pain, *xi*-cleft points of three *yang* meridians of foot, i.e. ST34, GB36 and BL63, as well as *xi*-cleft point of foot-*taiyin* meridian, SP8. (2) Selecting acupoint of stomach meridian of foot-*yangming*, with excessive *qi* and blood, ST36, for *bi* syndrome of lower limbs, acting on promoting joint functions. (3) Selecting GB34, the influential point of tendon, acting on relaxing and strengthening tendons. (4) Selecting SP10, acting on controlling and replenishing blood, as well as promoting spleen and stomach functions in transportation and transformation, specially relieving postoperative nausea and vomiting. (5) Selecting empirical point, EX-LE29, being effective in postoperative analgesia [18]. Acupuncture is analgesic through pro-

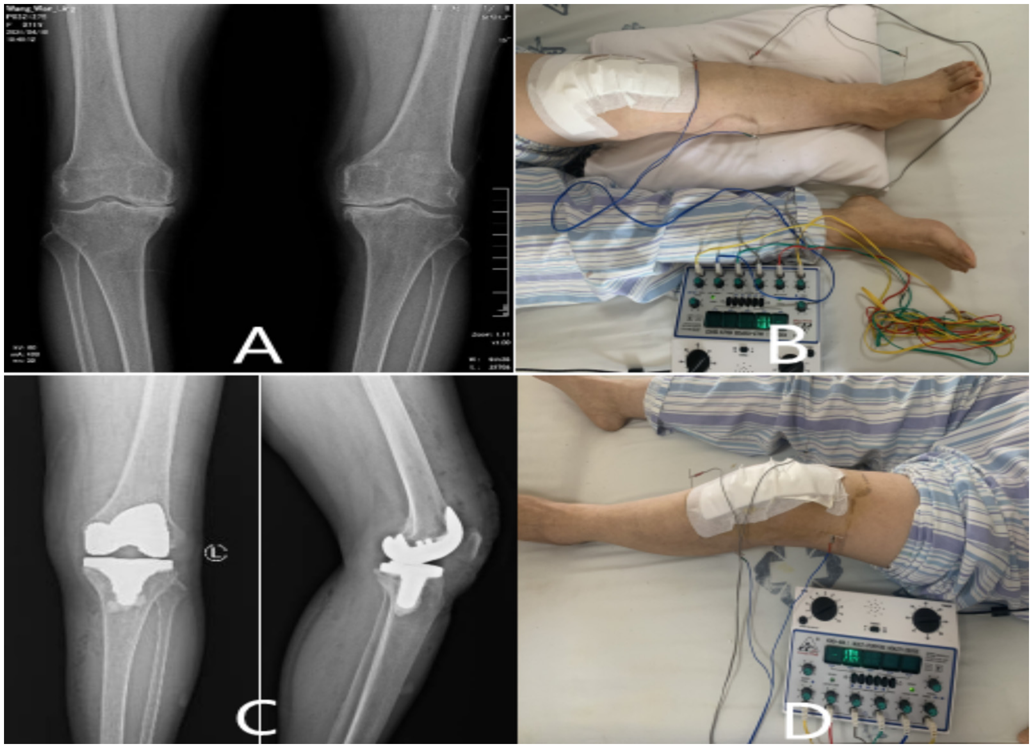

Fig. 3. Postoperative analgesiaby EA in patient with TKA.
Notes: A Standing orthotopic film of knee joint before operation. B EA on the day after operation. C Anterior-lateral film of knee joints on the 2nd day after operation, showing satisfactory placement of prosthesis. D EA on the 2nd day after operation.

moting circulation of meridian, regulating *yin* and *yang*, activating *qi* and blood circulation, strengthening the antipathogenic *qi* and eliminating pathogens to modulate the body [19]. It is what is called that "only due to the fact that there is sufficient *zhengqi* (antipathogenic *qi*) inside the body, that is why *xieqi* (pathogens) cannot invade the body". It implies the holism of TCM and it is crucial in acupuncture analgesia.

In recent years, many scholars have intended to clarify the effect mechanism of acupuncture analgesia from the perspective of molecular biology and neurobiology. Han [20,21] revealed in view of neurobiology that EA stimulation may be analgesic by regulating neurochemicals and he also discovered the differences in timeliness of analgesia between high-frequency electric stimulation and low-frequency one. In neurochemical mechanisms of EA analgesia, the release of neuropeptide is promoted by activating the opioid system to prevent from the conduction of nociceptive stimulation. Besides, small molecule neurotransmitters, e.g. serotonin, norepinephrine and acetylcholine, are all closely involved in the process of analgesia [22]. Zhu et al [23]. believed that the mechanism of acupuncture analgesia is divided into segmental and systemic mechanism. The selection of either distal points or nearby points is associated with the strength of acupuncture manipulation. Moreover, the neuromechanisms underlie the differences in the analgesic effect produced by these two selections of acupoints.

The results in this trial showed that the pain symptoms in the EA combination group were alleviated obviously in tendency based on the consecutive 3-day VAS assessment after operation as compared with the celecox group in patients. Besides, 24 h after operation, VAS score in the celecox group was obviously higher than the score before operation, while there was no difference in pain symptoms before and after operation in the EA combination group. The trauma in TKA is the chief reason of pain aggravation. Hence, the results above indicate that EA contributes to relieving postoperative acute pain of TKA. The study by Li et al [24]. confirmed that EA significantly relieved pain after TKA and postoperative painful reactions and reduced the risk of postoperative vertigo. SRSS score was improved gradually in two group. Compared with the celecox group, SRSS score was reduced obviously in the EA combination group. It is proved in the research [25] that acupuncture relieves the symptoms of insomnia through neuroelectrophysiological regulation. IL-6 and TNF-α are the inflammatory factors synthesized and released by macrophages, monocytes and endotheliocytes [26]. IL-10 is an anti-inflammatory factor. The expressions of IL-6 and TNF-α are conductive to inhibition, thus alleviating the inflammatory response [27]. TKA brings a huge stimulus in patients, thus activates the body's defense barrier to generate a large number of inflammatory factors and aggravate damage in the body. It has been shown in researches that IL-6 in serum is the optimal indicator to detect the strength of surgical stress, whose peak usually appears 24 h after operation [28,29]. This discovery is consistent with the conclusion in this trial. 48 and 72 h after operation, IL-6 and TNF-α in serum of the celecox group were obviously higher than those in the EA combination group and IL-10 was lower apparently than the EA combination group. It is suggested that EA promotes the release of IL-10 and reduces the generation of IL-6 and TNF-α, and then, alleviates inflammatory reaction induced by TKA. This conclusion is coincident with the previous researches [30,31]. It is believed that the stimulation with EA causes the alternation of stress factors in the body, thus induces analgesia. In addition, the relevant study has shown that the intervention with EA may effectively reduce postoperative cognitive dysfunction after TKA and lay the foundation for the objective and real evaluation of indicators after operation [32]. Therefore, it is confirmed that EA has a positive effect in analgesia after TKA. Specially, EA is significantly effective to alleviate pain at acute stress stage and a series of physiopathological responses induced.

In summary, EA therapy is conductive to relieving pain and promoting the early rehabilitation of knee joints after TKA. It is applicable in various postoperative analgesia. However, there is few clinical trial of EA in analgesia after TKA. Because of many meridians distributed in human body and diversity of acupoint selection, it is the difficulty to specify the acupoints for analgesia after TKA, thus, the exploration is required on this topic. Besides, the stimulation parameters of EA in postoperative analgesia have not been specified yet. It needs to further discuss and develop a specification on the objective parameters so as to better provide the theoretic reference for clinical practice. Finally, the authors believe that EA is an unique green therapy for analgesia and it is of more practical significance in postoperative analgesia and will better serve human health undertakings.

State of human and animal rights

The intervention conformed to the ethical criteria.

Declaration of Competing Interest

The authors declare that there is no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] Roth ML, Tripp DA, Harrison MH, Sullivan M, Carson P. Demographic and psychosocial predictors of acute perioperative pain for total Kenn arthroplasty. Pain Res Manag 2016;12(3):185–94.

[2] Bi Z, Wu WX. Advanced analgesic effect of preoperative femoral nerve block in elderly patients with total knee replacement under general anesthesia. Zhejiang J Trauma Surg 2016;21(6):1210–12.

[3] Ren XP, Cai LY, Zhang JN, Xu MY. Effect of ultrasound-guided single obturator nerve block on continuous femoral nerve block for postoperative analgesia of patients underwent total knee arthroplasty. Jiangsu Med J 2016;42(15):1674–6.

[4] Shen B, Weng XS, Liao R, Qu TB, Zhang CL, Cao L, et al. Expert consensus in enhanced recovery after total hip and knee arthroplasty in China: pain and sleep management. Chin J Bone Jt Surg 2016;9(2):91–7.

[5] Xu LL, Zhu HS. Clinical observation of the analgesic effect of electroacupuncture after proximal femoral nail antirotation operation for femoral intertrochanteric fracture. Shanghai J Acupunct Moxibustion 2018;37(5):557–9.

[6] Zhang WL. Clinical observation of early analgesia of electroacupuncture after total hip arthroplasty. Guangzhou: Guangzhou University of Chinese Medicine; 2019.

[7] Osteoporosis Group of Chinese Orthopaedic AssociationGuidelines for osteoarthritis (2018). Chin J Orthop 2018;38(12):705–15.

[8] Li YJ, Zhou Y. Electroacupuncture of different wave patterns for acute attack of lumbar disc herniation. World J Acupunct Moxibustion 2021;31(3):207–11.

[9] Chang Q, Chang GS, Zhao LY. Discovery and clinical application of Nèimádiăn (‘†‘…‘º‘»‘‚‘¹) point. World J Acupunct Moxibustion 2015;25(2):53–7.

[10] Liu RX, Yuan XL, Yu J, Quan Q, Meng HY, Wang C, et al. An updated meta-analysis of the asporin gene D-repeat in knee osteoarthritis: effects of gender and ethnicity. J Orthop Surg Res 2017;12(1):148.

[11] Law RJ, Nafees S, Hiscock J, Wynne C, Williams NH. A lifestyle management programme focused on exercise, diet and physiotherapy support for patients with hip or knee osteoarthritis and a body mass index over 35: a qualitative study. Musculoskelet Care 2019;17(1):145–51.

[12] Jesús CL, Araceli OR, Irene TS, Isabel LT, Ángel GVM, Carmen VM. Association between the functional impact of osteoarthritis and body mass index in women. Rev Med Chile 2018;146(9):987–93.

[13] Funck BT, Nethander M, Movérare SS, Richette P, Ohlsson C. Causal factors for knee, hip and hand osteoarthritis: a Mendelian randomization study in the UK Biobank. Arthritis Rheumatol 2019;71(10):1634–41.

[14] Khlopas H, Khlopas A, Samuel LT. Current concepts in osteoarthritis of the Ankle: review. Surg Technol Int 2019;35:280–94.

[15] Lin HB, Huang WH, Chen XH, Zhang GD, Yu ZX, Wu XW, et al. Digital design of internal fixation for distal femoral fractures via 3D printing and standard parts database. Natl Med J China 2016;96(5):344–8.

[16] Gao YQ, Jia Q, Xie S, Yin LW, Xue JX, Kou LH, et al. Clinical trials for thyroidectomy under acupuncture-aided anesthesia by using electroacupuncture or transcutaneous acupoint electrical stimulation of different acupoints. Acupunct Res 2017;42(4):332–7.

[17] Fang JQ. Acupuncture analgesia: panorama multi-dimensional treatment of pain. Acupunct Res 2018;4(8):459–66.

[18] Han XC, Xing QZ, Li Y, Dong X, Ma XW, Zhang L, et al. Clinical trial of electroacupuncture at Neimadian-point for cancer pain. Acupunct Res 2020;45(10):845–50.

[19] Wang Q, Bao YX. Analysis on the principle of acupuncture analgesia. China J Tradit Chin Med Pharm 2019;34(10):4911–13.

[20] Han JS. New evidence to substantiate the frequency specificity of acupuncture-induced analgesia. Acupunct Res 2001;26(3):224–7.

[21] Han JS. Induction of the release of central neuropeptides by peripheral electrical stimulation. J Peking Univ 2002;34(5):408–13.

[22] Chen WT, Fu GQ, Shen WD. Progresses of studies on acupuncture analgesia for postoperative reaction. Acupunct Res 2013;38(1):83–7.

[23] Zhu B, Rong PJ, Pen H, Gao XY, Li YQ. Study on the segmental mechanism and systemic mechanism of acupuncture analgesia. Acupunct Res 2007;32(3):153.

[24] Li Y, Wang P, Deng DH, Yan W, Li L, Jiang HJ. Electroacupuncture for pain control after total knee arthroplasty: a meta-analysis. Chin J Tissue Eng Res 2021;25(6):957–63.

[25] Ren ZQ, Yuan X, Zhao R, Xu JL. Acupuncture for insomnia attention deficit and the influence of nerve electrophysiology. World J Acupunct Moxibustion 2018;28(03):161–6.

[26] Jin L. Curative effect of manual massage and electroacupuncture on sciatica and its effect on serum IL-6, TNF-α and MCP-1. Mod J Integr Tradit Chin West Med 2017;26(19):2107–9.

[27] Turel MK, Kerolus M, Deutsch H. The role of minimally invasive spine surgery in the management of pyogenic spinal discitis. J Craniovertebr Junction Spine 2017;8(1):39–43.

[28] Xie JW, Ma J, Yao H, Yue C, Pei FX. Multiple boluses of intravenous tranexamic acid to reduce hidden blood loss after primary total knee arthroplasty without tourniquet: a randomized clinical trial. J Arthroplast 2016;31(11):2458–64.

[29] Zhang SY, Cao GR, Huang Q, Li YT, Xu H, Pei FX. Risk factors associated with interleukin 6 level in serum after total knee arthroplasty. Chin J Reparative Reconstr Surg 2018;32(8):1001–5.

[30] Liang Y, Zhou J, Sun J, Wang CY, Li BW, Zhou CL, et al. Professor Fang Jianqiao's experiences in clinical application of electroacupuncture in analgesia. Shanghai J Acupunct Moxibustion 2016;35(12):1387–9.

[31] Sun N, Zhou MT. Analgesic effect of electroacupuncture after surgery for breast cancer and its influence on TNF-α, IL-6 and immune function. Shanghai J Acupunct Moxibustion 2019;38(12):1350–4.

[32] Zhao FY, Zhang ZY, Zhao YX, et al. The effect of electroacupuncture preconditioning on cognitive impairments following knee replacement among elderly: a randomized controlled trial. World J Acupunct Moxibustion 2018;28(4):231–6.